

Phagocytic Overload and Lysosome-Dependent Cell Death Down-Regulation in Frontotemporal Dementia Patients

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Introduction

Frontotemporal Dementia (FTD) is a deadly neurodegenerative disorder afflicting upwards of 22 people in every 100,000 people worldwide. Patients with FTD see a buildup of TAU and TDP-43 proteins in their brain, resulting in damage to neurons; this can cause a number of symptoms which can be categorized as behavioral changes and speech or movement impediments. While neither treatment nor prevention exists for FTD, attempts to slow disease progression down cost patients on average \$120,000 a year, with behavioral variant FTD (bvFTD) and late stage FTD costing more. This study aims to explore pathways with differential expression potentially responsible for the protein buildup exhibited in bvFTD patients to make therapies more effective and affordable.

Methods

For this study, we selected 80 bvFTD patient and 80 control samples from GEO's GSE140830 dataset, which consists of peripheral blood gene expression data. Using GEO2R, we performed differential expression analysis and a t-test and exported the findings into Google Sheets. From there, we sorted by significant ($p\text{-value} \leq 0.05$) and large LogFC ($|\log\text{FC}| \geq 0.3$). The resulting genes were then inserted into String-DB, which directed us to a list of enriched KEGG pathways and the associated genes. Through KEGG pathways and GeneCards we pursued these pathways, specifically the Necroptosis and Phagosome pathways, understanding the ramifications these genes may have on pathways and its connection to FTD.

Results

The Necroptosis pathway (false discovery rate of $1.9 \cdot 10^{-3}$) contained four relevant genes that increase lysosomal membrane permeabilization HSP90AA1 (LogFC: -0.418), BIRC3 (LogFC: -0.296), STAT1 (LogFC: -0.324), and EIF2AK2 (LogFC: -0.312). This suggests an inhibition of lysosome dependent cell death [2]. The Phagosome pathway (false discovery of $9.8 \cdot 10^{-3}$) revealed three relevant genes: ATP6V1G1 (LogFC: -0.407), reducing the capability to acidify phagosomes; ACTG1 (LogFC: 0.459) indicating high phagocytic load, which reduces phagosome maturation rate [3]; FCGR3B (LogFC: -0.482), reducing the specificity and efficiency of phagosome fusion with lysosomes.

Discussion

Our results provide a potential explanation for the mechanism by which misfolded protein build up occurs in FTD patients. While the cause of protein misfolding requires further investigation, we propose phagocytic overload and downregulation of lysosome induced cell death as potential explanations for lack of effective immune response against misfolded proteins. As Tau proteins are naturally present in healthy patients, perhaps appropriate immune response is critical to handle accidental misfolding before misfolding proliferates. These results suggest targeted gene-therapy focused on facilitating phagocytosis may slow disease progression and reduce symptoms, possibly at a more effective and affordable rate than current treatments.

References

1. Wang, F et. al. *Traffic*. **2018**; 19: 918–931
2. Liebl, David et. al, *Journal of Cell Science*. **2009**